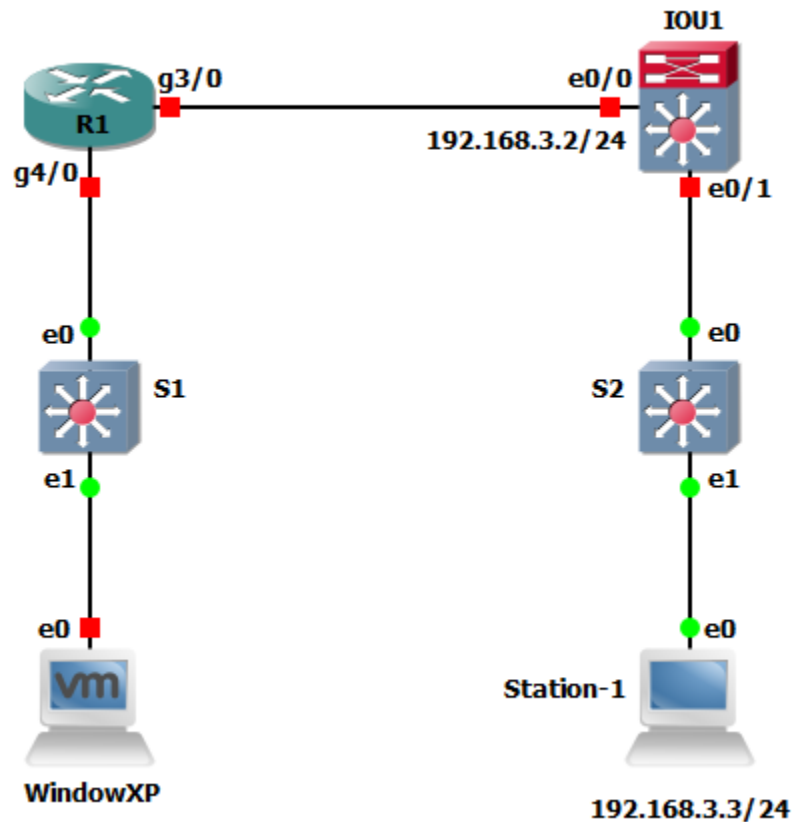


Lab: How to configure SNMPv3 (DAYUTH THY)

323



PC: with PowerSNMP Free Manager

Configuration on Router (Agent side):

```
!=====
ip access-list standard PERMIT-SNMP
permit 192.168.1.0 0.0.0.255
Exit
snmp-server engineID local 0000000003
snmp-server group SNMP-G1 v3 priv write SNMP-RW access PERMIT-SNMP
snmp-server view SNMP-RW iso included
snmp-server host 192.168.1.3 version 3 priv SNMP-Admin
snmp-server user SNMP-Admin SNMP-G1 v3 auth sha authpass priv aes 128 pripass
snmp-server enable traps
End
!=====
R2(config)#snmp-server group SNMP-G1 v3 ?
```

Configure SNMPv3 to send trap to PowerSNMP Free Manager

auth group using the authNoPriv Security Level
noauth group using the noAuthNoPriv Security Level
priv group using SNMPv3 authPriv security level

Configure SNMPv3 to send trap to PowerSNMP Free Manager

SNMP-Admin: is the username

SNMP-G1: is the group name

authpass: is the password for authentication

privpass: is the password for encryption

!Verify

```
R2#sh run | s snmp
mmi snmp-timeout 180
snmp-server engineID local 0000000003
snmp-server group SNMP-G1 v3 priv write SNMP-RW access PERMIT-SNMP
snmp-server view SNMP-RW iso included
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps ds1
snmp-server enable traps tty
snmp-server enable traps eigrp
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps xgcp
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop conf
ig
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service
-up
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps diameter
snmp-server enable traps auth-framework sec-violation auth-fail
snmp-server enable traps trustsec-sxp conn-srcaddr-err msg-parse-err conn-config
-err binding-err conn-up conn-down binding-expn-fail oper-nodeid-change binding-
conflict
snmp-server enable traps aaa_server
snmp-server enable traps atm subif
snmp-server enable traps bfd
snmp-server enable traps bgp cbgp2
snmp-server enable traps bstun
```

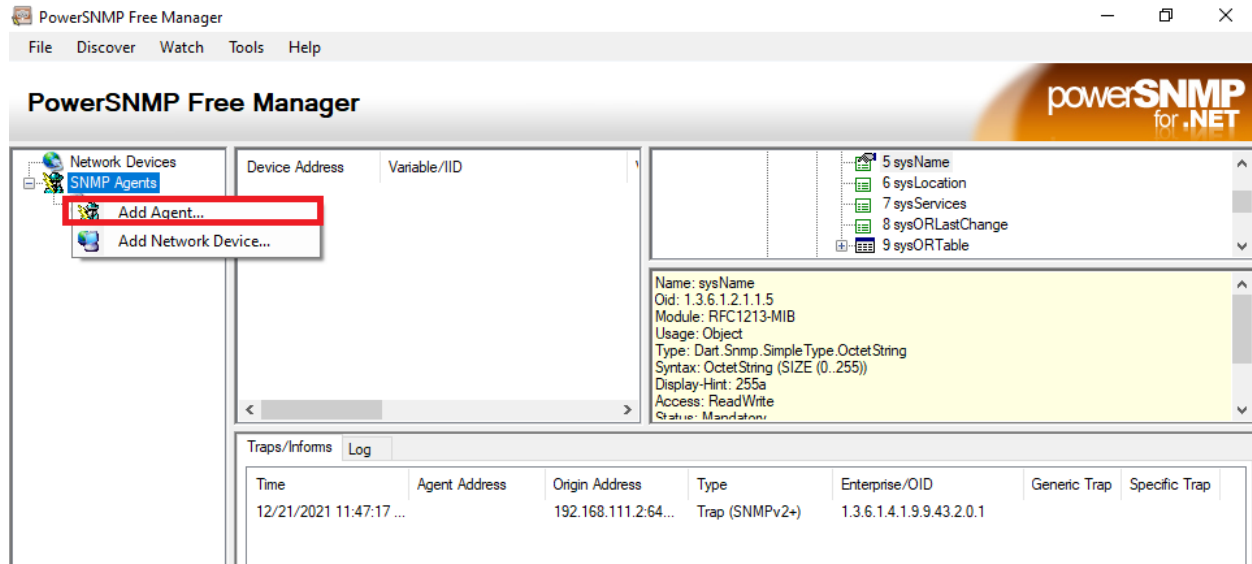
Configure SNMPv3 to send trap to PowerSNMP Free Manager

```
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps memory bufferpeak
snmp-server enable traps cnpd
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps dial
snmp-server enable traps dlsr
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps entity-ext
snmp-server enable traps entity
snmp-server enable traps fru-ctrl
snmp-server enable traps resource-policy
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps hsrp
snmp-server enable traps ipmobile
snmp-server enable traps ipmulticast
snmp-server enable traps isis
snmp-server enable traps mempool
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps pppoe
snmp-server enable traps cpu threshold
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps rsvp
snmp-server enable traps ipsla
snmp-server enable traps stun
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps pw vc
snmp-server enable traps waas
snmp-server enable traps event-manager
snmp-server enable traps firewall serverstatus
```

Configure SNMPv3 to send trap to PowerSNMP Free Manager

```
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pki
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps rf
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps mpls vpn
snmp-server enable traps voice
snmp-server enable traps dnis
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server host 192.168.111.3 version 3 priv SNMP-Admin
How to configure on PC:
```

Configure SNMPv3 to send trap to PowerSNMP Free Manager



Right-click on **SNMP Agents** > **Add Agent...**

The 'Agent Configuration' dialog box is shown. It contains the following fields and options:

- Address: 192.168.111.2
- Port: 161
- Version: ☐ 1 ☐ 2 ☒ 3
- Community: public
- User Authentication and Privacy (v3) section:
 - Name: SNMP-Admin
 - Auth Password: authpass
 - Priv Password: pripass
 - Auth Protocol: Sha
 - Priv Protocol: Aes128

NOTE: SHA and TripleDes are compliant with FIPS-140 standards.

Buttons: OK, Cancel

Address: 192.168.111.2 is the IP address of SNMP agent installed

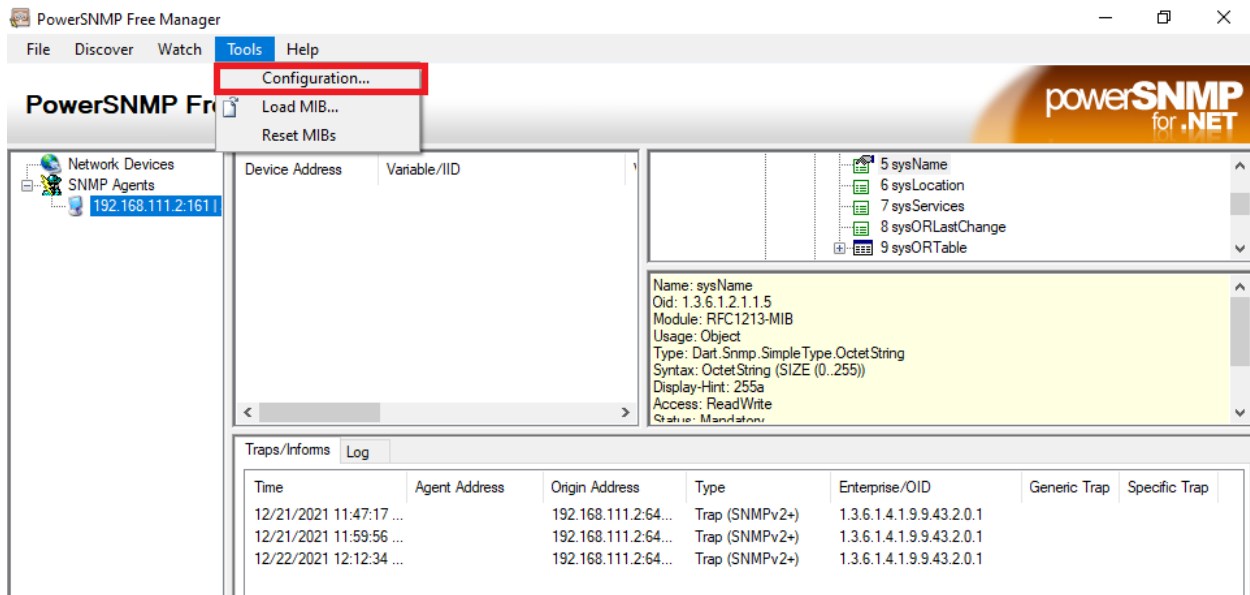
Port: 161, Version: 3

Name: is username

Auth Password: is the password of Authentication, Auth Protol: according to router

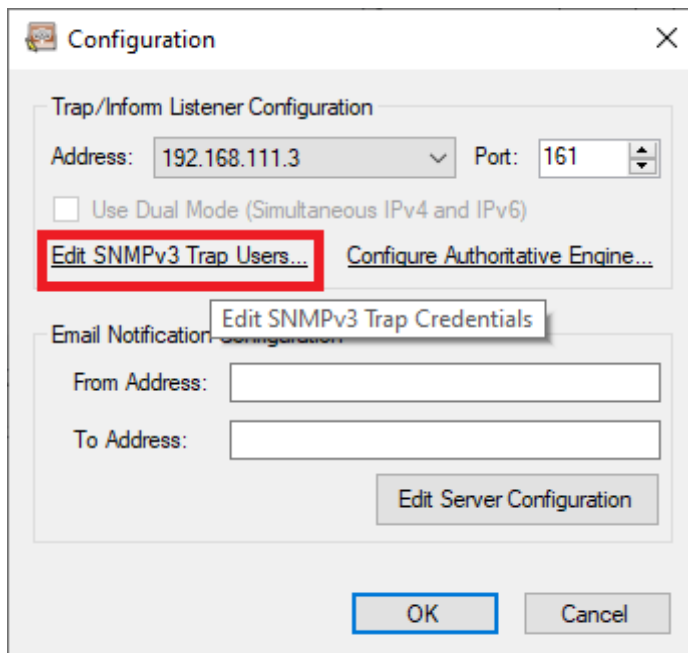
Priv Password: is the password of encryption, Priv Protocol: match to router configuration

Configure SNMPv3 to send trap to PowerSNMP Free Manager



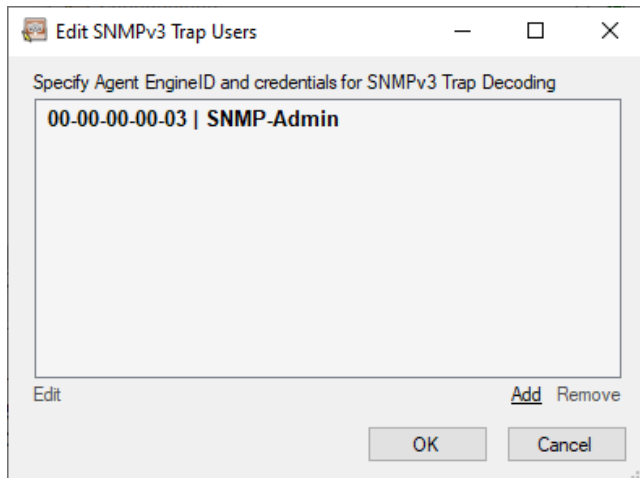
To configure traps:

Click **Tool > Configuration**



Click on **Edit SNMPv3 Trap Users...**

Configure SNMPv3 to send trap to PowerSNMP Free Manager



Click **Add**

Engin ID from command: **snmp-server engineID local 0000000003**

Name: is the username

Auth Password: is the password of Authentication,

Priv Password: is the password of encryption

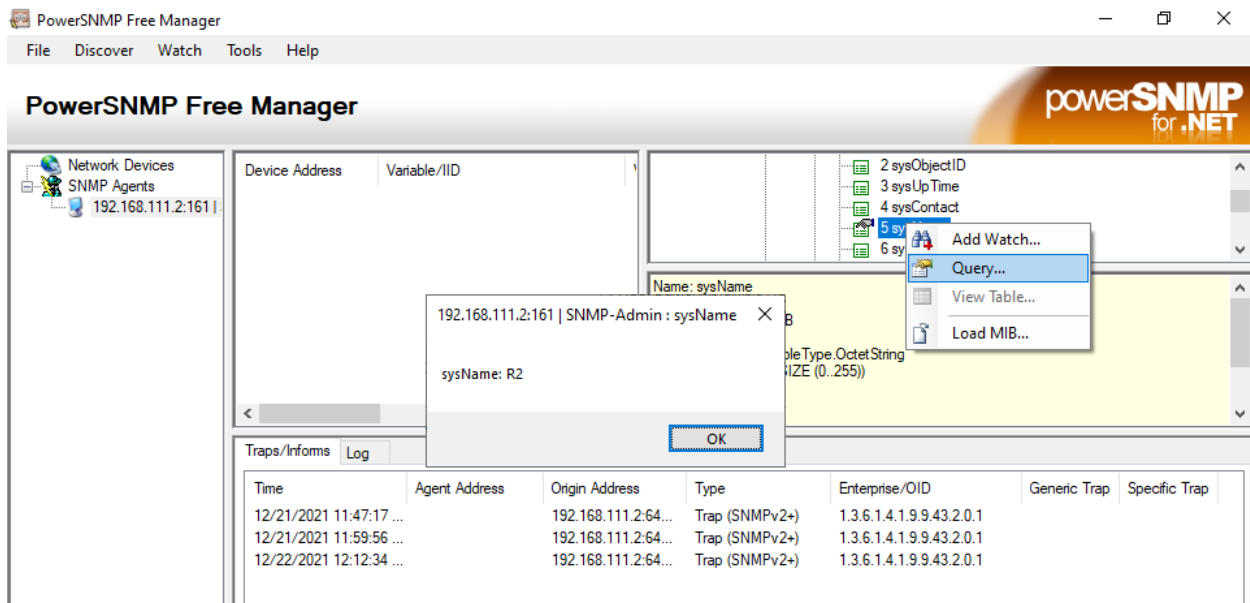
Auth Protocl: according to router

Priv Protocl: match to router configuration

Address: is IP of PC to listen to trap, Port change to 192 and click **OK**

Configure SNMPv3 to send trap to PowerSNMP Free Manager

To test:



Right click on **SysName** > **Query...**

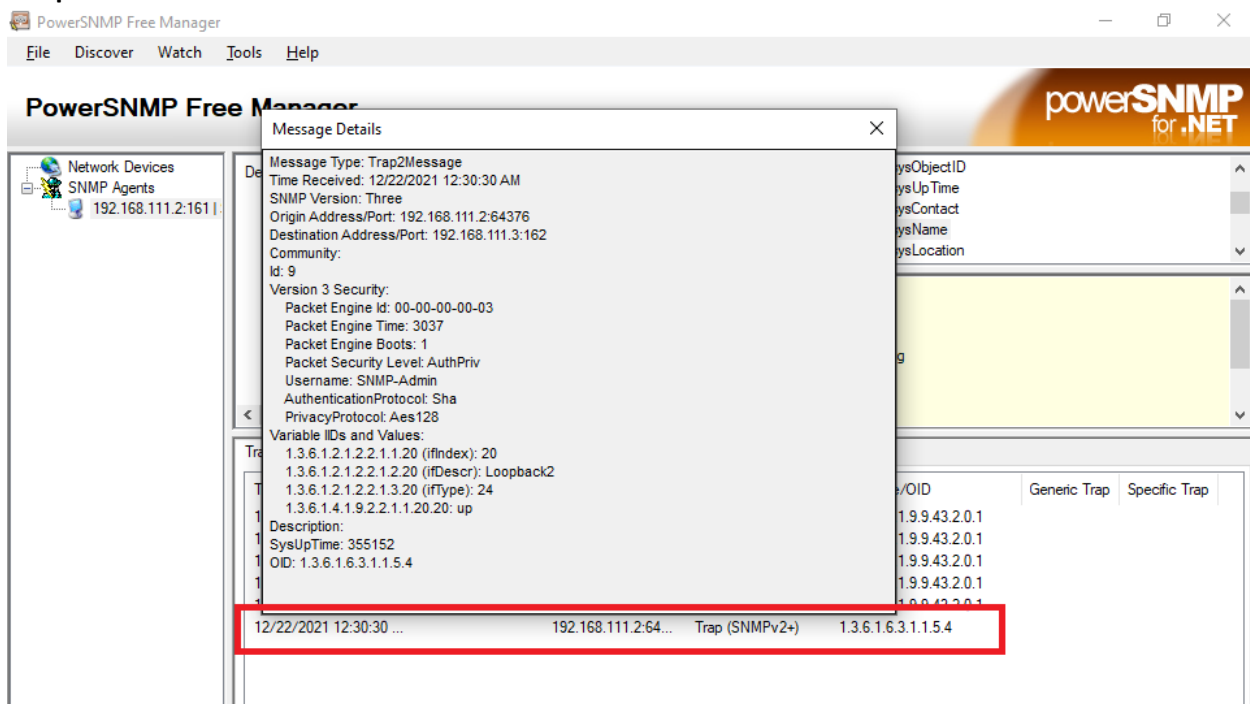
The result of the query is R2, which is the name of the router

Add a loopback interface on R2 by using command below:

R2(config)#interface lo2

R2(config-if)#

***Dec 22 08:30:33.395: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback2, changed state to up**



Double click on the Trap message under the Traps/Informs Tab of the bottom display pane, it show the Loopback2 is up state

Configure SNMPv3 to send trap to PowerSNMP Free Manager

Verify if SNMP traps has been sent from Router:

R2#sh snmp

Chassis: 2048003

14 SNMP packets input

0 Bad SNMP version errors

0 Unknown community name

0 Illegal operation for community name supplied

0 Encoding errors

6 Number of requested variables

0 Number of altered variables

6 Get-request PDUs

0 Get-next PDUs

0 Set-request PDUs

0 Input queue packet drops (Maximum queue size 1000)

20 SNMP packets output

0 Too big errors (Maximum packet size 1500)

0 No such name errors

0 Bad values errors

0 General errors

0 Response PDUs

6 Trap PDUs

SNMP Dispatcher:

queue 0/75 (current/max), 0 dropped

SNMP Engine:

queue 0/1000 (current/max), 0 dropped

0 Unknown Security Models

0 SNMP Invalid Messages

1 SNMP Unknown PDU handlers

0 Unsupported Security Level

4 Unknown User Names

3 Unknown EngineIDs

0 Not In Time Windows

0 Wrong MD5 or SHA Digests

0 Decryption Errors

6 messages have been sent

SNMP logging: enabled

Logging to 192.168.111.3.162, 0/10, 6 sent, 0 dropped.

R2#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#in

R2(config)#interface lo

R2(config)#interface lo2

R2(config-if)#sh

R2(config-if)#no

Configure SNMPv3 to send trap to PowerSNMP Free Manager

*Dec 22 08:38:11.443: %LINK-5-CHANGED: Interface Loopback2, changed state to administratively down

*Dec 22 08:38:12.448: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback2, changed state to down

R2(config-if)#no sh

R2(config-if)#end

R2#

R2#sh snmp

*Dec 22 08:38:17.117: %SYS-5-CONFIG_I: Configured from console by console

R2#sh snmp

Chassis: 2048003

14 SNMP packets input

0 Bad SNMP version errors

0 Unknown community name

0 Illegal operation for community name supplied

0 Encoding errors

6 Number of requested variables

0 Number of altered variables

6 Get-request PDUs

0 Get-next PDUs

0 Set-request PDUs

0 Input queue packet drops (Maximum queue size 1000)

23 SNMP packets output

0 Too big errors (Maximum packet size 1500)

0 No such name errors

0 Bad values errors

0 General errors

0 Response PDUs

9 Trap PDUs

SNMP Dispatcher:

queue 0/75 (current/max), 0 dropped

SNMP Engine:

queue 0/1000 (current/max), 0 dropped

0 Unknown Security Models

0 SNMP Invalid Messages

1 SNMP Unknown PDU handlers

0 Unsupported Security Level

4 Unknown User Names

3 Unknown EngineIDs

0 Not In Time Windows

0 Wrong MD5 or SHA Digests

0 Decryption Errors

SNMP logging: enabled

Logging to 192.168.111.3.162, 0/10, 10 sent, 0 dropped.



10 messages have been sent after sh and no shut has been issued on Loopback 2 interface

Configure SNMPv3 to send trap to PowerSNMP Free Manager

R2#

*Dec 22 08:38:18.067: %LINK-3-UPDOWN: Interface Loopback2, changed state to up

*Dec 22 08:38:19.072: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback2, changed state to up

Link to download: <https://powersnmp-free-manager.software.informer.com/download/>